

Candidate Name: _____

Class Index No

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Promotional Examination 2008 Pre-university 2

BIOLOGY HIGHER 1 PAPER 1 MULTIPLE CHOICE

8875/1

Thursday

11 September

1h

Additional materials:
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and admission number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

- 1 The Golgi apparatus has two faces – the cis face and the trans face. Which of the following correctly describe the two faces of the Golgi apparatus.

	Cis face	Trans face
A	Usually faces away from the endoplasmic reticulum	Usually located near the endoplasmic reticulum.
B	Receives vesicles transported from the endoplasmic reticulum.	Vesicles are pinched off and transported to other sites.
C	Receives cis fats from other organelles.	Receives trans fats from other organelles.
D	Fuses with the endoplasmic reticulum.	Fuses with the nuclear membrane.

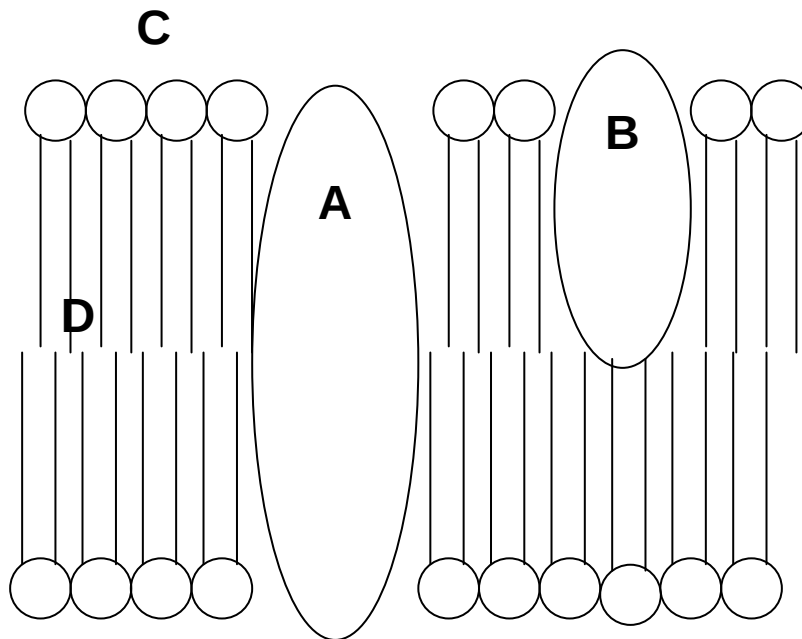
- 2 Pancreatic cells are placed in warm isotonic saline solution to which radioactively labelled amino acids were added.
At intervals after adding the amino acids, samples of the cells were removed, sections cut and the sites of radioactivity determined.
Which one of the following represents the order in which radioactivity appeared in organelles?

A	Rough endoplasmic reticulum	Smooth endoplasmic reticulum	Golgi apparatus	Secretory vesicles
B	Rough endoplasmic reticulum	Smooth endoplasmic reticulum	Secretory vesicles	Golgi apparatus
C	Smooth endoplasmic reticulum	Rough endoplasmic reticulum	Golgi apparatus	Secretory vesicles
D	Rough endoplasmic reticulum	Golgi apparatus	Smooth endoplasmic reticulum	Secretory vesicles

- 3 Which of the following events is responsible for the autodigestion of a cell?

- A** Degradation of the chromosomes in the nucleus.
- B** Activation of the enzymes in the cytoplasm.
- C** Leakage of the contents of the lysosomes.
- D** Excessive synthesis of digestive enzymes in the rough endoplasmic reticulum.

- 4 The list below shows the composition of various membranes. Which of the following would you expect to be the most viscous?
- A Phospholipids with saturated hydrocarbon tail and cholesterol
 - B Phospholipids with saturated hydrocarbon tail only
 - C Phospholipids with cis-unsaturated hydrocarbon tail only
 - D Phospholipids with trans-unsaturated hydrocarbon tail and cholesterol
- 5 The diagram below represents the cell surface membrane. In which region will you expect to find cholesterol?



- 6 Where in the nucleus would you expect a high concentration of rRNA?
- A nucleolus
 - B heterochromatin
 - C nuclear envelope
 - D nuclear pore

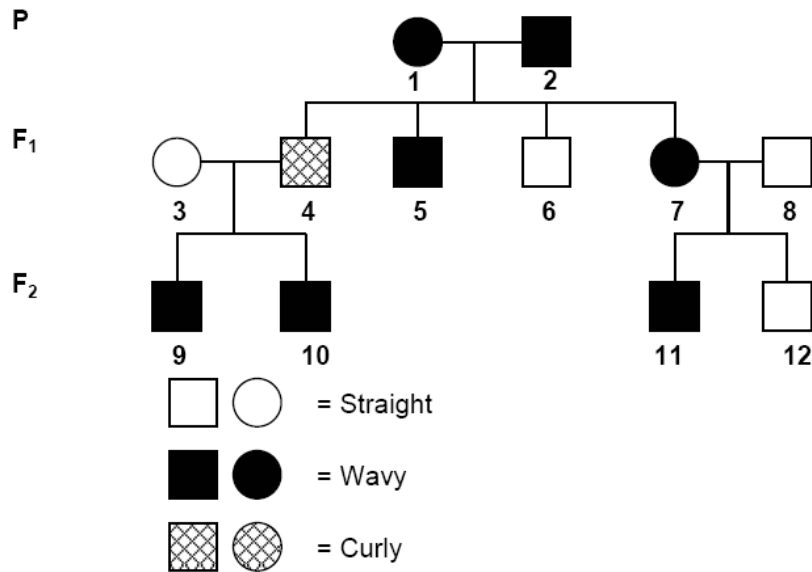
- 7 Which one of the following correctly illustrates the difference between RNA and DNA in a eukaryotic cell?

	DNA	RNA
A	Single stranded	Double stranded
B	Pentose	Hexose
C	Ribose	Deoxyribose
D	Thymine	Uracil

- 8 The potentiality for replication of DNA depends on
- A hydrogen bonds between base and sugar groups
 - B high energy bonds between base and phosphate groups.
 - C hydrogen bonds between bases.
 - D covalent bonds between bases.
- 9 The complementary messenger RNA triplet for the DNA triplet GAT would read
- A TAG
 - B CUG
 - C CTA
 - D CUA
- 10 Which of the following statements concerning heterochromatin is false?
- A Heterochromatin stains deeply with DNA-binding dyes
 - B Heterochromatin contains a relatively large number of genes
 - C Heterochromatin is found in the telomere and the centromere
 - D Heterochromatin remains relatively condensed throughout the cell cycle and condenses early in cell division.

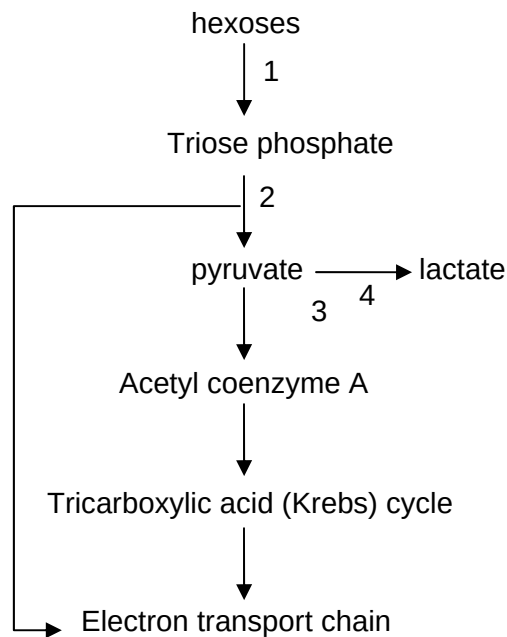
- 11 In which of the following regions would you expect the DNA to be the least condensely packed?
- A Centromere
 - B Euchromatin
 - C Heterochromatin
 - D Telomere
- 12 What is the purpose of attaching the poly-A tail and the capping of the 5' end with a modified guanine nucleotide?
- I facilitate the export of mature mRNA from the nucleus
 - II protect the mRNA from degradation by hydrolytic enzymes
 - III allow the anticodon on the tRNA to form hydrogen bonds with the codon on the mRNA.
 - IV help ribosomes attach to the 5' end of the mRNA
- A I, II and III
 - B II, III and IV
 - C I, II and IV
 - D I, III and IV
- 13 Which one of the following is not a characteristic of enhancers?
- A They are located away from the gene they control.
 - B They are made up of control elements that can bind to transcriptional activators.
 - C They are made up of control elements that can bind to transcriptional repressors.
 - D Each enhancer is made up of only one control element.
- 14 Control of gene expressions in eukaryotes includes all of the following **except**
- A feedback inhibition of enzyme activity by allosteric control
 - B DNA methylation
 - C transcription factors
 - D alternative splicing of RNA transcripts

Refer to the pedigree chart below for questions 30 and 31.



- 15** Which of the following statements is true?
- A** Wavy hair is dominant
 - B** Individual 1 is homozygous
 - C** Individual 4 is homozygous
 - D** Curly hair is recessive to straight hair
- 16** Which one of the following statements has a 50% chance of being true?
- A** One of the parents of individual 2 had the same phenotype as individual 2.
 - B** If individual 10 married someone with wavy hair, the first child would have wavy hair.
 - C** If individual 6 married someone with straight hair, the first offspring would have straight hair.
 - D** If individual 5 married someone with curly hair, the offspring will have straight hair.

- 17 The diagram below summarises the stages involved in respiratory metabolism of mammalian skeletal muscle.



Stage 4 in the diagram occurs in mammalian muscle cells under anaerobic conditions. Why is stage 4 necessary?

- A to detoxify pyruvate
 - B to form lactate which is immediately reconverted to glycogen
 - C to provide reduced nicotinamide adenine dinucleotide for the electron transport chain
 - D to oxidise the reduced nicotinamide adenine dinucleotide facilitating stage 2
- 18 What is the precise role of oxygen in cellular respiration?
- A To oxidise reduced NAD^+ in glycolysis
 - B To combine with the carbon in glucose to form carbon dioxide
 - C To act as the final hydrogen acceptor in the electron transport chain
 - D As an oxidising agent in the stages in Krebs cycle that involve oxidation
- 19 Where would you expect the enzymes involved in glycolysis to be located?

- A In the fluid matrix of the cytoplasm
- B In the mitochondrial matrix
- C In the nuclear sap
- D On the smooth endoplasmic reticulum

- 20 An aqueous suspension of isolated chloroplasts will evolve oxygen if illuminated in the presence of a certain type of compound and which colours of light are required for maximum oxygen evolution?

	Type of compound	Colours of light at which maximum oxygen evolution occur
A	Electron acceptor	green
B	Electron acceptor	Blue and red
C	Electron donor	Green
D	Electron donor	Blue and red

- 21 Which one of the following represents an homologous evolutionary feature?

- A Wings of a bat and flippers of a dolphin
- B Wings of a pigeon and wings of a dragonfly
- C Legs of a cat and legs of an ant
- D Eye of a snail and eye of a rabbit

- 22 A bacterium synthesises restriction enzymes to digest DNA from infecting viruses. Which one of the following is a mechanism to prevent the restriction enzymes from digesting the bacterium's own DNA?

- A The addition of methyl groups to adenines or cytosines within the sequences recognised by the restriction enzymes.
- B Extensive supercoiling of the DNA molecule to prevent the access of the enzyme to the restriction sites.
- C Coiling of the DNA molecule around proteins
- D Synthesis of competitive inhibitors of the restriction enzymes

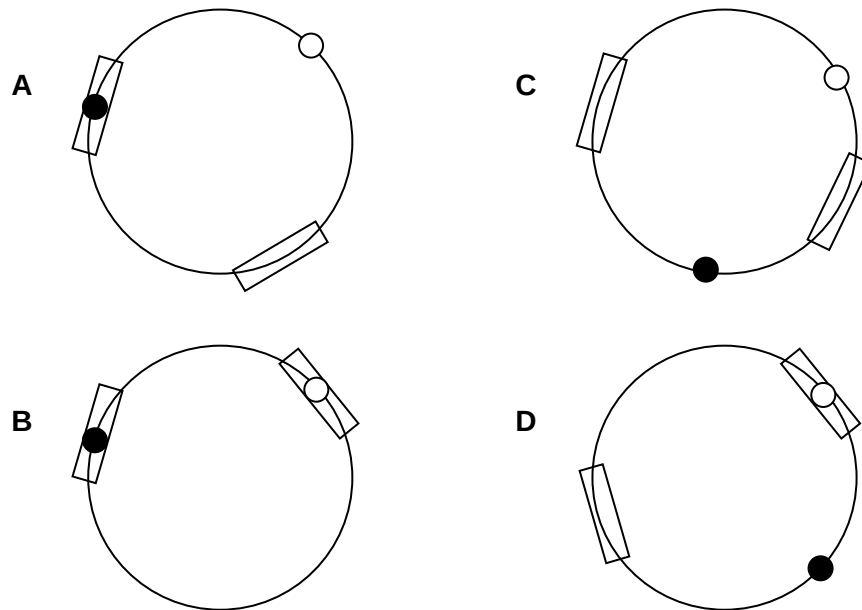
- 23 The restriction enzyme **AluI** is used to digest a plasmid into which a gene of interest is to be inserted. This digestion results in blunt ends on the plasmid.

Which of the following restriction enzymes can be used to digest the DNA molecule from which the gene of interest that is to be inserted into the above plasmid is obtained?

	Restriction enzyme	Restriction site
A	BamHI	5' ...G G A T C C... 3' 3' ...C C T A G G... 5'
B	EcoRI	5' ...G A A T T C... 3' 3' ...C T T A A G... 5'
C	HaeIII	5' ...G G C C... 3' 3' ...C C G G... 5'
D	HindIII	5' ...A A G C T T... 3' 3' ...T T C G A A... 5'

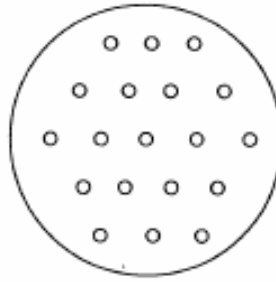
- 24 Plasmids are used in gene cloning technology as a cloning vector. Which one of the following is a reason why plasmids are suitable to be used as a vector?
- A It is easy to insert foreign DNA *in vitro* to form recombinant plasmids
 - B Plasmids can be replicated *in vitro*
 - C Plasmids are smaller than nucleoids
 - D The plasmid DNA has a similar arrangement to that of a eukaryotic cell's chromosomal DNA.

- 25 Which one of the following plasmids is suitable to be used in gene cloning?

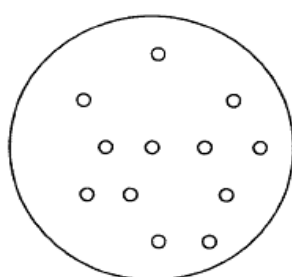


Legend: ○ Restriction site
 □ Selectable markers
 ● Origin of replication

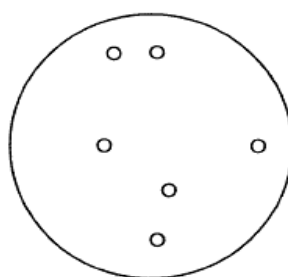
and tetracycline. The plasmids were used to transform *E. coli* and the bacteria grown on a nutrient medium. The resulting master plate is shown in the diagram.



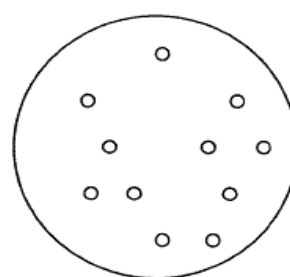
Transformed cells were selected by replica plating the bacteria colonies onto media containing various antibiotics. The results of the replica plating are shown in the diagrams below.



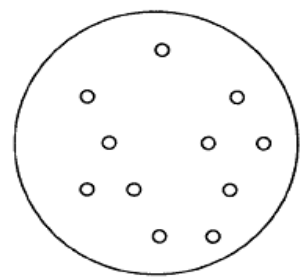
medium with ampicillin and streptomycin



medium with the antibiotic kanamycin

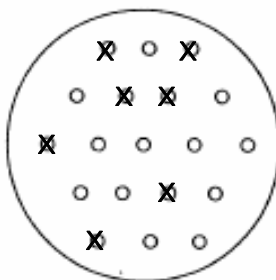


medium with ampicillin, streptomycin and tetracycline

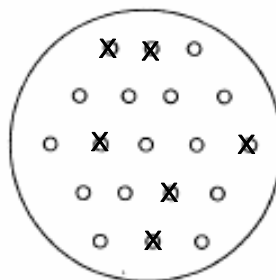


medium with tetracycline

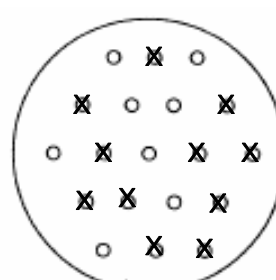
Which one of the following correctly marks out the colonies that contain the recombinant plasmids?



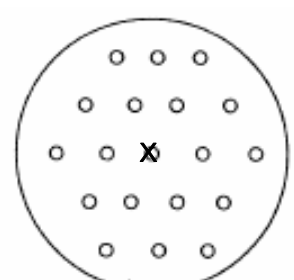
A



B



C



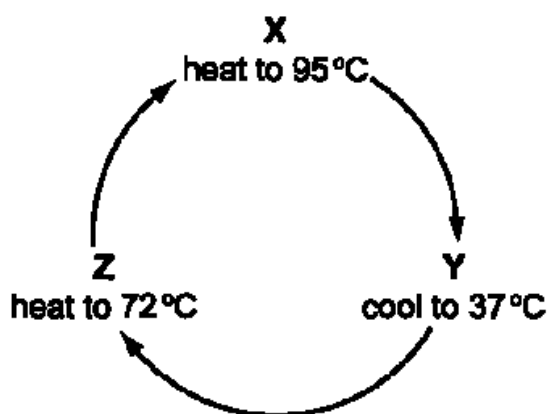
D

- 27 Which one of the following is **not** required in a mixture for polymerase chain reaction?

[Turn over

- A Special primers
- B Restriction enzymes
- C Nucleotides
- D DNA polymerase

- 28 The diagram shows the changes in temperature of the reaction mixture during the polymerase chain reaction (PCR).



Which combination correctly matches the event in PCR with the temperature in the reaction mixture?

	X	Y	Z
A	Separation of DNA strands	Annealing of RNA primers	DNA synthesis
B	Separation of DNA strands	DNA synthesis	Annealing of RNA primers
C	DNA synthesis	Separation of DNA strands	Annealing of RNA primers
D	Annealing of RNA primers	DNA synthesis	Separation of DNA strands

- 29 Embryonic stem cells offer great promise to the cure of many diseases because
- A they are totipotent.
 - B they will differentiate and produce cells which are normal.
 - C they are more readily available than other cells.
 - D they can be easily cultured in vitro.

- 30 Which one of the following is **not** a function of a stem cell?

- A Differentiate into cardiac muscle cells in the heart
- B Produce new red blood cells to replace worn out ones
- C Synthesise insulin when the pancreatic cells are damaged
- D Produce bone cells during growth

END OF PAPER

BLANK PAGE

- 2 A
- 3 C
- 4 B
- 5 D
- 6 A
- 7 D
- 8 C
- 9 D
- 10 B
- 11 B
- 12 C
- 13 D
- 14 A
- 15 C
- 16 B
- 17 D
- 18 C
- 19 A
- 20 B
- 21 A
- 22 D
- 23 C
- 24 A
- 25 D
- 26 D
- 27 B
- 28 A
- 29 B
- 30 C